Chapter 7: Project Cost Management

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations

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Learning Objectives (1 of 2)

- Develop a justification for project cost management and its importance in achieving project success
- Explain basic project cost management principles, concepts, and terms
- Describe the process of planning cost management
- Discuss different types of cost estimates and methods for preparing them
- Using an example of an information technology (IT) project, list and describe the processes of determining a budget and preparing a cost estimate

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Learning Objectives (2 of 2)

- Justify the use of earned value management and project portfolio management to assist in cost control
- Describe how project management software can assist in project cost management
- Discuss considerations for agile/adaptive environments

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The Importance of Project Cost Management

- IT projects have a poor track record for meeting budget goals
- Cost is a resource sacrificed to achieve a specific objective, or something given up in exchange
  - Usually measured in monetary units that must be paid to acquire goods and services
- Cost overrun is the additional percentage or monetary amount by which actual costs exceed estimates
  - May be caused by unclear project requirements, involving new technology or business processes
- Benefit is the value received from something, e.g. revenue from sales, ability to do something you could not perform before, removing defects
  - These may be converted into a currency value and compared to the cost

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Basic Principles of Cost Management (1 of 2)

- Types of costs and benefits
  - Tangible costs or benefits are those costs or benefits that an organization can easily measure in monetary terms
  - Intangible costs or benefits are costs or benefits that are difficult to measure in monetary terms
    - E.g. Costs not billed to the project
    - E.g. Goodwill, prestige
  - Direct costs are costs that can be directly related to producing the products and services of the project
    - E.g. Salaries, hardware/software cost
    - Project managers should focus on direct costs because they can be controlled
  - Indirect costs are costs that are not directly related to the products or services of the project, but are indirectly related to performing work on the project
    - E.g. electricity, water supply, paper towels
  - Sunk cost is money that has been spent in the past; consider it gone like a sunken ship that can never be raised
    - When deciding what projects to invest in or continue, you should not include sunk costs

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Basic Principles of Cost Management (2 of 2)

- Learning curve theory states that when many items are produced repetitively, the unit cost of those items decreases in a regular pattern as more units are produced
- Reserves are monetary amount included in a cost estimate to mitigate cost risk by allowing for future situations that are difficult to predict
  - Contingency reserves allow for future situations or identified risks that may be partially planned for (sometimes called known unknowns) and are included in the project cost baseline
    - E.g. If an organization knows it has 20% rate of turnover for IT personnel, it should include contingency reserves for recruiting and training costs of IT personnel
  - Management reserves allow for future situations or unidentified risks that are unpredictable (sometimes called unknown unknowns)
    - E.g. If a project manager gets sick or a supplier goes out of business, management reserves could be set aside for the resulting costs
    - Linked to an organization's policy and a project's level of uncertainty, available with higher management approval, and not included in the cost baseline

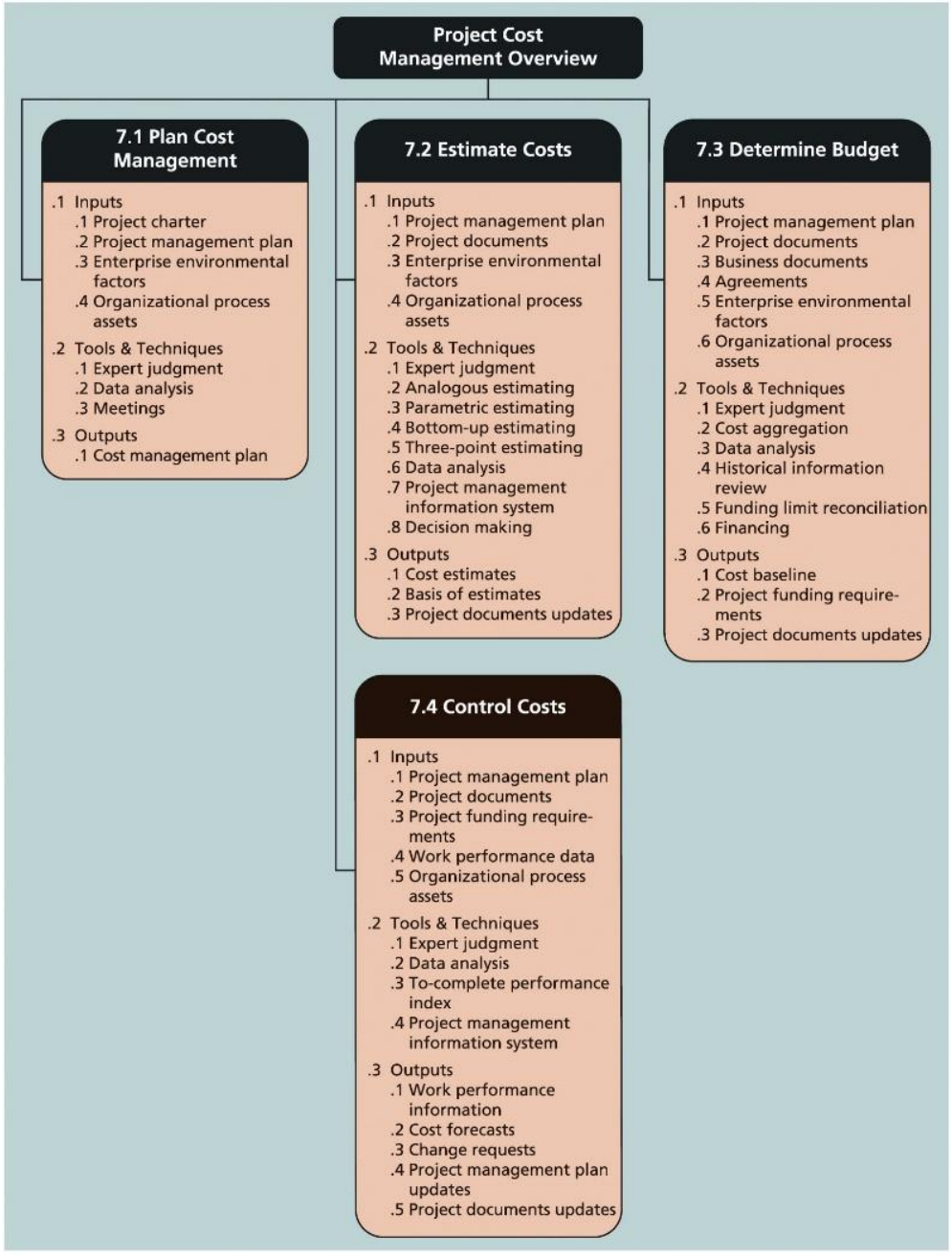
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Project Cost Management Processes

- Required to ensure that the project is completed within an approved budget

1. Planning cost management
   - Determining the policies, procedures, and documentation that will be used for planning, executing, and controlling project cost
2. Estimating costs
   - Developing an estimate of the costs of the resources needed to complete a project
3. Determining the budget
   - Allocating the overall cost estimate to individual work items to establish a baseline for measuring performance
4. Controlling costs
   - Controlling changes to the project budget

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FIGURE 7-1 Project cost management overview

1. Planning Cost Management

- The first step in project cost management is planning how the costs will be managed throughout the life of the project
  - The project team uses expert judgment, analytical techniques, and meetings to develop the cost management plan
- Cost management plan includes:
  - Level of accuracy
    - E.g. rounding guideline (e.g. to the nearest $100), contingency funds (e.g. 10-20%)
  - Units of measure
    - E.g. Labor hours or days
  - Organizational procedure links
    - A selected WBS component becomes a point at which the organization tracks and controls project costs in its accounting system; this point is called a control account
    - It allows management to ask: For this portion of the WBS, how much did we plan, how much work have we accomplished, and how much have we actually spent?
  - Control thresholds
    - Amount of variation allowed before actions need to be taken, e.g. 10% of baseline cost
  - Rules of performance measurement
    - How often actual costs will be tracked and to what level of detail
  - Reporting formats
    - Format and frequency of cost reports
  - Process descriptions
    - How to perform all of the cost management processes

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2. Estimating Costs

- Project managers must take cost estimates seriously if they want to complete projects within budget constraints
  - Types of cost estimates
  - Tools and techniques for estimating costs
  - Typical problems associated with IT cost estimates

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Types of Cost Estimates (1 of 2)

Type of Estimate	When Done	Why Done	Typical Range of accuracy
Rough order of magnitude (ROM)	Very early in the project life cycle, often 3+ years before project completion	Provides estimate of cost for selection decisions	-50% to + 100%
Budgetary	Early, 1–2 years before project completion	Puts amount in the budget plans	-10% to +25%
Definitive	Later in the project, 1 year or less before project completion	Provides details for making purchasing decisions, estimates actual costs	-5% to +10%

Table 7-1 Types of cost estimates

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Types of Cost Estimates (2 of 2)

- Estimates are usually done at various stages of a project
  - Should become more accurate as time progresses
- It is important to provide basis for the estimate, i.e. supporting details for estimates
  - Assumptions used in creating the estimate, e.g. labor rate, estimated effort
  - Cost estimation tools and techniques used, e.g. analogous method
- A large percentage of total project costs are often labor costs
  - FTE or full-time equivalent staff is a standard way to refer to the total staffing required regardless of whether people work full time or part time
    - E.g. when an employer has a 40-hour workweek, employees who are scheduled to work 40 hours per week are 1.0 FTEs. Employees scheduled to work 20 hours per week are 0.5 FTEs
  - Labor costs are often higher for contractors, so it is important to distinguish between internal and external resources

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Cost Estimation Tools and Techniques

- Expert judgment
- Analogous
  - Use the actual cost of a previous, similar project as the basis for estimating the cost of the current project
- Bottom-up estimates
  - Involve estimating individual work items or activities and summing them to get a project total (see later)
- Three-point estimates
  - Involve estimating the most likely, optimistic, and pessimistic costs for items
- Parametric estimating
  - Uses project characteristics (parameters) in a mathematical model to estimate project costs (see later)

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Typical Problems with IT Cost Estimates Inaccuracy

- Estimates are done too quickly
  - Many estimates must be done quickly before clear system requirements have been produced
  - Estimates are done at various stages, and project managers need to explain the rationale for each estimate
- People lack estimating experience
  - Not enough accurate reliable project data on which to base estimates are available
  - Organizations should keep reliable project information including estimates
- Human beings are biased toward underestimation
  - Senior project managers might make estimates based on their own abilities and forget that younger people will be working on the project
- Management desires accuracy
  - Management might ask for an estimate but really want a more accurate, and lower, number
  - Project managers must use leadership and negotiation skills to stand by the estimates

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How to Develop a Cost Estimate and Basis of Estimates (1 of 3)

- One of the best ways is to study sample cost estimate
- Before creating an estimate,
  - Gather as much information as possible about the project
  - Ask how the organization plans to use the cost estimate
    - E.g. use definitive estimate if it is for contract award or cost performance reporting
  - Clarify the ground rules and assumptions
    - Documented in the basis of estimates
    - See attachment document



- Using the basis of estimates, a cost estimate is developed
  - Use more than one approach to estimate (costs by approaches A and B) and see how the estimates differ

$$percent\ difference\ of\ cost = \frac{|costA - costB|}{average(costA, costB)} * 100$$

  - Also, have the cost estimate approved

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How to Develop a Cost Estimate and Basis of Estimates (2 of 3)

Surveyor Pro Project Cost Estimate Created October 5

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 2 Totals	% of Total
WBS Items					
1. Project Management				**$306,300**	**20%**
Project manager	960	$100	$96,000		
Project team members	1,920	$75	$144,000		
Contractors (10% of software development and testing)			$66,300		
2. Hardware				**$76,000**	**5%**
2.1 Handheld devices	100	$600	$60,000		
2.2 Servers	4	$4,000	$16,000		
3. Software				**$614,000**	**40%**
3.1 Licensed software	100	$200	$20,000		
3.2 Software development*			$594,000		
4. Testing (10% of total hardware and software costs)			$69,000	**$69,000**	**5%**
5. Training and Support				**$202,400**	**13%**
Trainee cost	100	$500	$50,000		
Travel cost	12	$700	$8,400		
Project team members	1,920	$75	$144,000		
Subtotal			$1,267,700		
6. Reserves (20% of total estimate)		0.2*1,267,700 →	$253,540	**$253,540**	**17%**
Total project cost estimate				**$1,521,240**	

*See software development estimate.

FIGURE 7-2 Surveyor Pro project cost estimate

How to Develop a Cost Estimate and Basis of Estimates (3 of 3)

Surveyor Pro Software Development Estimate Created October 5

1. Labor Estimate	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	Calculations
Contractor labor estimate	3,000	$150	$450,000	3,000 * 150
Project team member estimate	1,920	$75	$144,000	1,920 * 75
Total labor estimate			**$594,000**	Sum above two values
2. Function point estimate	Quantity	Conversion Factor	Function Points	**Calculations**
External inputs	10	4	40	10 * 4
External interface files	3	7	21	3 * 7
External outputs	4	5	20	4 * 5
External queries	6	4	24	6 * 4
Logical internal tables	7	10	70	7 * 10
Total function points			**175**	Sum above function point values
Total labor hours (27 hours/function point)*			4,725	27*175
Cost/labor hour ($120/hour)			$120	Assumed value from budget expert
Total function point estimate			**$567,000**	4,725 * 120

* Based on historical data

FIGURE 7-3 Surveyor pro software development estimate

Development estimates differ by 4.65%

Labor Estimate by Problem Decomposition (1 of 2)

Expert judgment / analogous, bottom-up estimates

- Decompose scope statement into functions (WBS items by product) that can each be estimated individually, i.e. in line of code (LOC) or effort (person-month)

LOC-based example

The mechanical CAD software will accept two- and three-dimensional geometric data from a designer. The designer will interact and control the CAD system through a user interface. All geometric data and other supporting information will be maintained in a CAD database. Design analysis modules will be developed to produce the required output, which will be displayed on a variety of devices. The software will be designed to control and interact with peripheral devices that include a touchpad, scanner, laser printer, and large-bed digital plotter.

Function	Estimated LOC
User-interface and control facilities (UICF)	2,300
Two-Dimensional geometric analysis (2DGA)	5,300
Three-Dimensional geometric analysis (3DGA)	6,800
Database management (DBM)	3,350
Computer graphics display facilities (GCDF)	4,950
Peripheral control function (PCF)	2,100
Design analysis modules (DAM)	8,400
Estimated lines of code	33,200

R. S. Pressman and B. R. Maxim, Software Engineering A Practitioner's Approach, 9th edition, 2020.

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Labor Estimate by Problem Decomposition (2 of 2)

- Estimated LOC of the project is 33,200
- Average productivity for systems of this type is 620 LOC/pm
  - 1 pm (or a person-month) is the amount of time (effort) one person spends working on the software development project for one month [COCOMO II]
- Estimated effort = 33,200 LOC / 620 LOC per pm = 53.5 ~ 54 pm
- Average labor rate is $8000 per month
- Estimated labor cost = 54 pm * $8,000 per month = $432,000.

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Labor Estimate by Process Decomposition (1 of 2)

Expert judgment / analogous, bottom-up estimates

- Decompose the process into tasks to be performed (WBS items by process)
- May also decompose scope statement into problem functions, and have the tasks performed on each function
- Then, estimate the effort (person-months) required to do each task for each function

Activity →	CC	Planning	Risk analysis	Engineering		Construction release		Totals
Task →				Analysis	Design	Code	Test	
Function								
↓								
UICF				0.50	2.50	0.40	5.00	8.40
2DGA				0.75	4.00	0.60	2.00	7.35
3DGA				0.50	4.00	1.00	3.00	8.50
CGDF				0.50	3.00	1.00	1.50	6.00
DBM				0.50	3.00	0.75	1.50	5.75
PCF				0.25	2.00	0.50	1.50	4.25
DAM				0.50	2.00	0.50	2.00	5.00
Totals	0.25	0.25	0.25	3.50	20.50	4.50	16.50	46.00
% effort	1%	1%	1%	8%	45%	10%	36%	

R. S. Pressman and B. R. Maxim, Software Engineering A Practitioner's Approach, 9th edition, 2020.

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Labor Estimate by Process Decomposition (2 of 2)

- Estimated effort for the project is 46 person-months
- Average labor rate is $8000 per month
- Estimated labor cost = 46 pm * $8000 per month = $368,000

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Labor Estimate by Story Points for Agile Development (2 of 2)

Expert judgment / analogous, bottom-up estimates

ID	User Story	Reason for relative size	SP
US01	As a customer, I want to **register an account** so that I can make reservations.	Basic form, validation, database update	3
US02	As a customer, I want to **log in** so that I can access my account.	Reference story	3
US03	As a customer, I want to **search for available rooms by dates** so that I can find a suitable room.	Date and availability logic, database queries	8
US04	As a customer, I want to **view room details** so that I can choose a room.	Mainly data retrieval and display	2
US05	As a customer, I want to **make a room reservation** so that the room is reserved for my stay.	Availability checking, validation, multiple updates	8
US06	As a customer, I want to **pay online** so that my reservation is confirmed.	External payment gateway, security, error handling	13
US07	As a customer, I want to **view and cancel my reservations** so that I can manage my bookings.	Retrieval plus cancellation rules and updates	5
US08	As hotel staff, I want to **manage room information** so that customers see current room information.	CRUD operations and validation	5
	Total		**47 SP**

- Make an estimate for a user story by story point
  - Establish a reference story: The team first selects a reasonably understood story and assigns it a reference value, for example 3 SP
  - Compare other stories relatively: Compared with the reference story, how large and difficult is this story?
  - A common approach is Planning Poker with a modified Fibonacci scale: 1, 2, 3, 5, 8, 13, 20, 40, 100
- However, story point information may not yet be available during early planning of the project

R. S. Pressman and B. R. Maxim, Software Engineering A Practitioner's Approach, 9th edition, 2020.

Labor Estimate by Story Points for Agile Development (2 of 2)

- Estimated total story points for the project is 47 story points.
- Team velocity is 12 story points per increment (or sprint for Scrum).
- No. of increments required for the project = 47 story points / 12 story points = 3.92 ~ 4 increments
- An increment length is 2 weeks = 2 increments per month
- Estimated duration = 4 increments / 2 increments per month = 2 months
- No. of agile team members is 5 members.
- Average labor rate per person is $8,000 per month.
- Estimated labor cost = 5 members * $8,000 per month * 2 months = $80,000

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Function Point Analysis (FPA)

- Purpose is to measure and provide functional size of software to clients and stakeholders
  - Logical size, as opposed to physical size like line of code or objects
- Function points (FP) are a unit measure for software
  - To compare program requirements, design, or final program size independent of chosen design and implementation details
  - To estimate the size of the final program by using a specific conversion table between Function Points and program code size (LOC per FP)
  - To estimate project effort (FP per person-month)
- First proposed in 1979 by Allan Albrecht and has become a standard for measuring software size (ISO 20926:2009)
- Currently maintained by IFPUG (International Function Point User Group https://ifpug.org/),

D. Longstreet, Function Point Analysis Training Course
https://www.cs.helsinki.fi/u/taina/ohtu/fp.html
https://www.softwaremetrics.com/fpafund.htm
https://www.javatpoint.com/software-engineering-functional-point-fp-analysis



IFPUG
International Function Point USERS GROUP

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Function Point Measurement

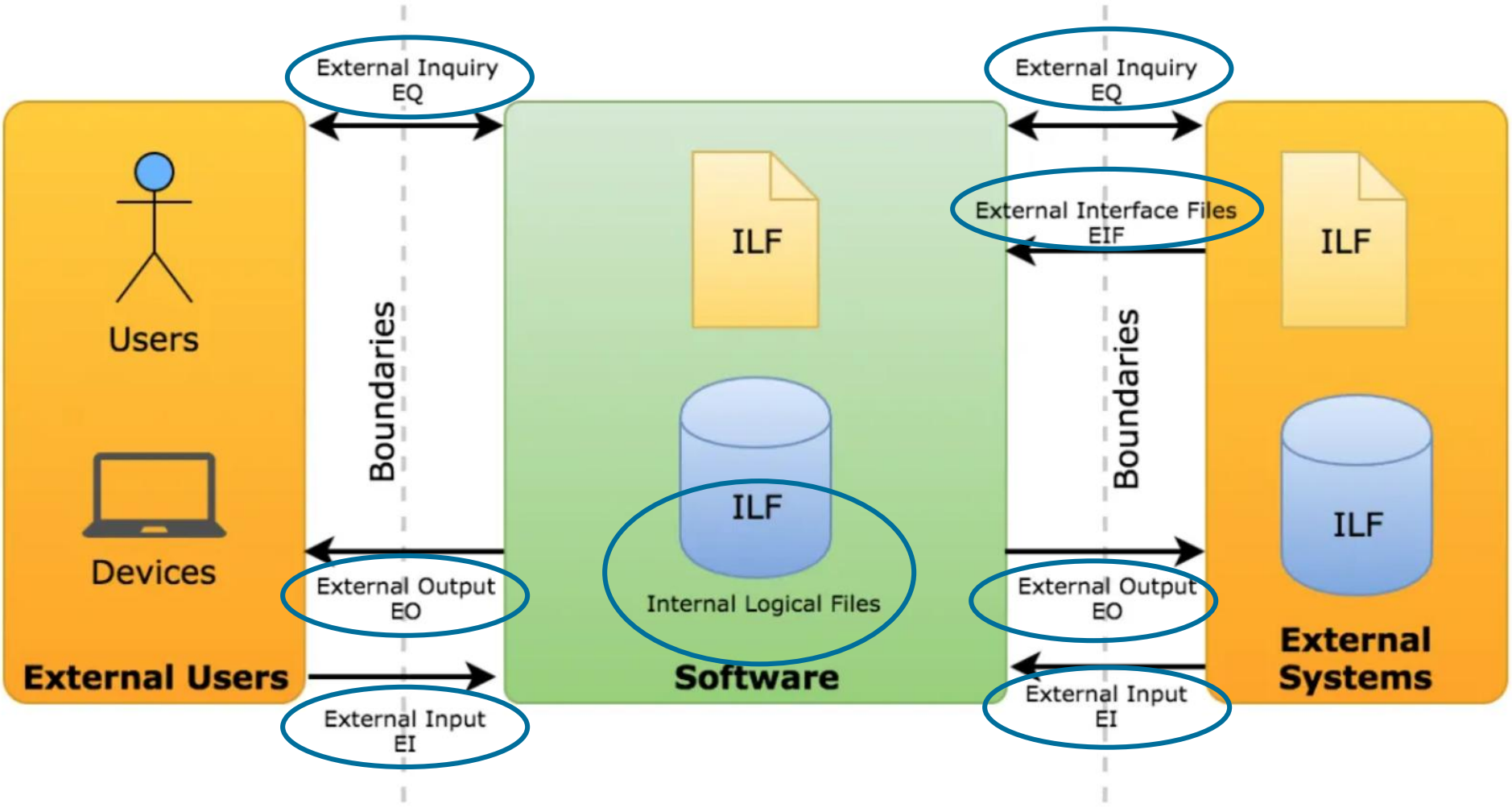
- A Function Point calculation is based on empirical evidence from a set of finished projects (parametric estimating)
  - Although the projects were from 70's, the results of the study are still surprisingly accurate
- Function Points are always an estimation
  - Different people may get the somewhat different FP values of the same problem domain or software, and this is not a serious issue
  - Even an estimation is better than nothing at all
  - FP allows for early estimations before a single line of code has been written

https://www.cs.helsinki.fi/u/taina/ohtu/fp.html

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Five Components of FPA (1 of 5)

Transaction (or data in motion) = EI, EO, EQ
Data (or data at rest) = ILF, EIF



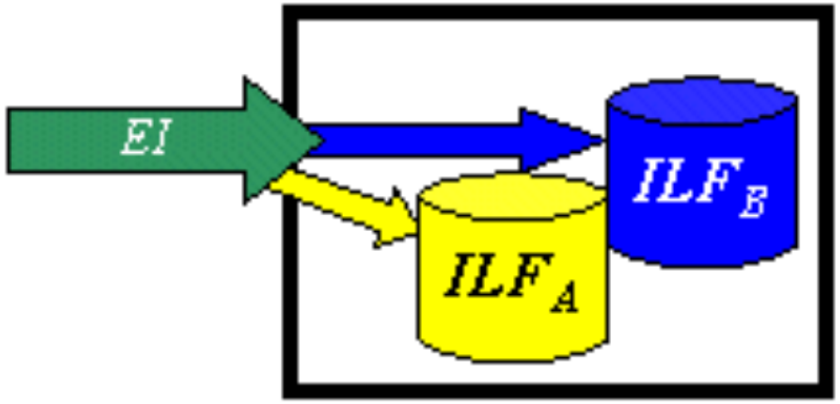
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https://melsatar.blog/2018/01/15/5-steps-to-software-development-effort-estimation/

Five Components of FPA (2 of 5)

External Inputs (EI) - is an elementary process in which data crosses the boundary from outside to inside. This data may come from a data input screen or another application. The data may be used to maintain one or more internal logical files. The data can be either control information or business information. If the data is control information it does not have to update an internal logical file. The graphic represents a simple EI that updates 2 ILF's (FTR's).

ILF = Internal Logical File
FTR = File Type Referenced



1 EI, 2 FTRs

- User provides data in an online application form, inputs text, selects options or rule-based data (e.g. check box, radio button), inputs control data (e.g. Add, Change, Delete buttons), uploads files, or feeds data into the system.
- An external application sends data to be stored/processed or both.
- Receive data from external sources as stream feeds at intervals.
- Receive a device event such as alarms or triggers with relevant data to be processed.

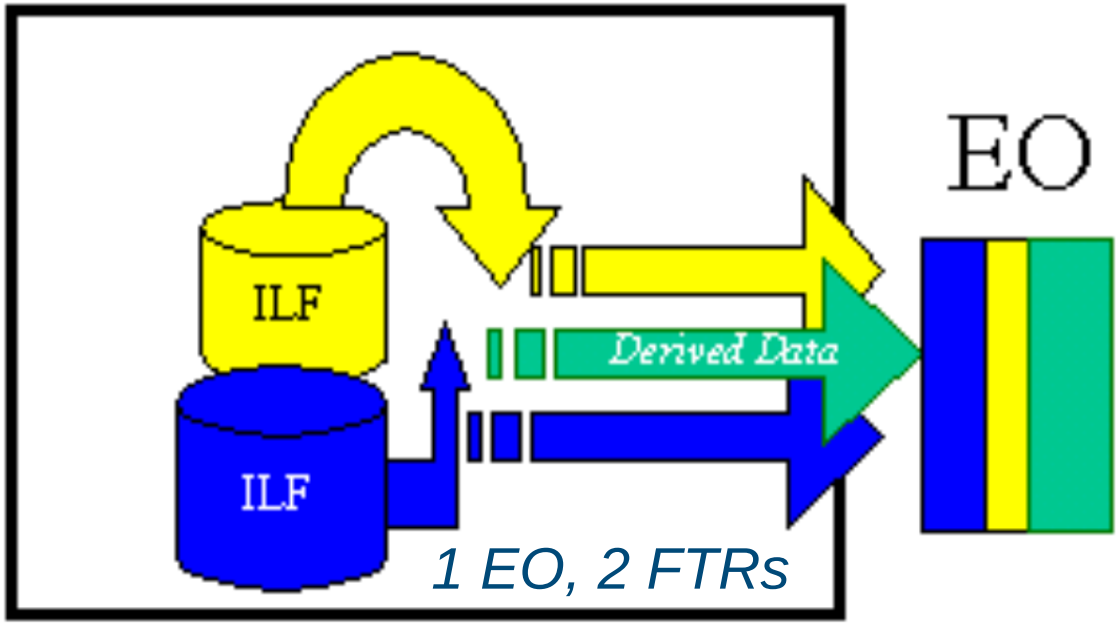
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D. Longstreet, Function Point Analysis Training Course
https://www.softwaremetrics.com/fpafund.htm

Five Components of FPA (3 of 5)

External Outputs (EO) - an elementary process in which derived data passes across the boundary from inside to outside. Additionally, an EO may update an ILF. The data creates reports or output files sent to other applications. These reports and files are created from one or more internal logical files and external interface file. The following graphic represents on EO with 2 FTR's there is derived information (green) that has been derived from the ILF's

ILF = Internal Logical File
FTR = File Type Referenced



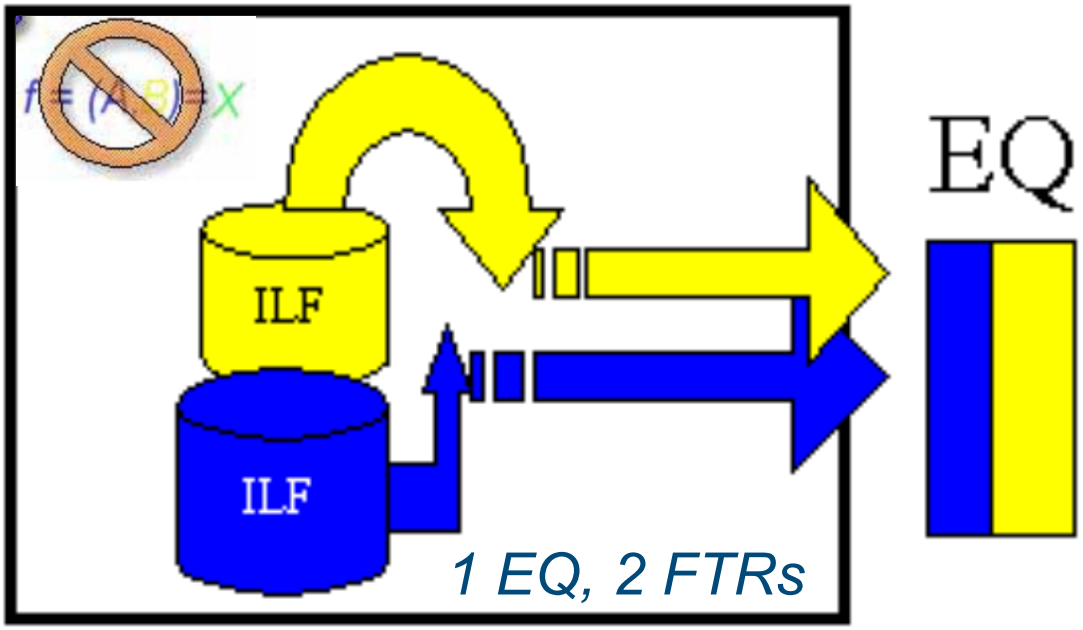
- Report, display result, print result, or output file with data fetched from the data source and some additional logic is applied to transform the data; derived data displayed in textual and graphical format
- Notification message as a result of some business logic processing, e.g. trading application may notify a broker that the customer trying to place an order does not have adequate funds in their account.
- An EO may update an ILF and may have the input side.

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Five Components of FPA (4 of 5)

External Inquiry (EQ) - an elementary process with both input and output components that result in data retrieval from one or more internal logical files and external interface files. The input process does not update any Internal Logical Files, and the output side does not contain derived data. The graphic below represents an EQ with two ILF's and no derived data.



ILF = Internal Logical File
FTR = File Type Referenced

- Query, search
- Report, display result, print result, or output file with data fetched from the data source and no data is updated or transformed.
- A display of a drop down list
- A reset (or restore) of all the modified fields to their saved values

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Five Components of FPA (5 of 5)

Internal Logical Files (ILF's) - a user identifiable group of logically related data that resides entirely within the applications boundary and is maintained through external inputs.

- ILF contains business data, control data, and rule-based data.
- File, local database, table, data source, or any other type of persistence mechanism

External Interface Files (EIF's) - a user identifiable group of logically related data that is used for reference purposes only. The data resides entirely outside the application and is maintained by another application. The external interface file is an internal logical file for another application.

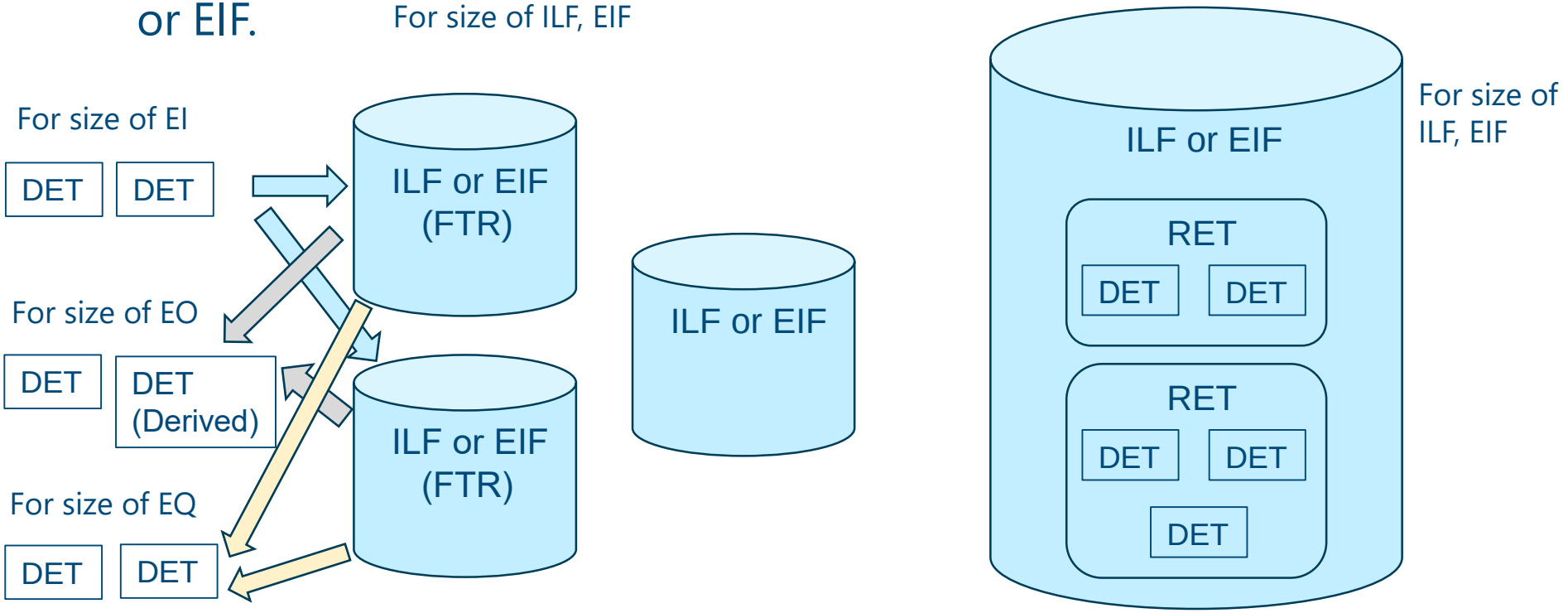
Data transmitted or persisted outside the application boundary, which can be accessed through the use of a remote application interface, a remote server, or data which may be maintained by another application and managed through remote interface.

D. Longstreet, Function Point Analysis Training Course.
https://www.softwaremetrics.com/fpafund.htm
https://sourceforge.net/p/functionpoints/wiki/Function%20Point%20Analysis/

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Measuring Function Size by Counting DET, FTR, and RET

- Data Element Type (DET) – data element — For size of EI, EO, EQ, ILF, EIF
- File Type Referenced (FTR) – ILF or EIF referenced by a transaction (EI, EO, EQ) — For size of EI, EO, EQ
- Record Element Type (RET) – subgroup of data elements within an ILF or EIF. — For size of ILF, EIF



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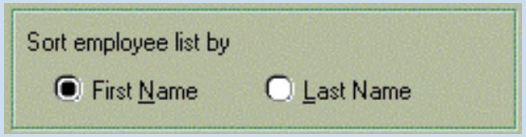
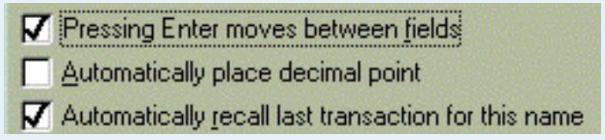
Counting DET to Measure Function Size

- Data Element Type (DET) For size of EI, EO, EQ, ILF, EIF
  - Unique user recognizable, non-recursive, non-repetitive field. A DET is a dynamic field, read from a file or created from other DETs. A DET can invoke transactions
  - E.g. Data field (numeric, text, photograph, sound), computed value, control button, radio button, checkbox, error message, confirmation message

D. Longstreet, Function Point Analysis Training Course.

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Counting DET for GUI

Data	Count	Example
Radio buttons	Count 1 only for all radio buttons in the frame	
Check boxes	Count 1 for each check box in the frame	
Drop down list	Count 1	
Action key (Command buttons)	Count 1 for the action it invokes	Add, Change, Delete, Inquire, OK buttons
Multiple action keys that perform the same function	Count 1 for the action it invokes	
Display of a graphical image and icon	Count 1	
Sound bytes	Count 1	
Photographic image	Count 1	
Error message	Count 1	“Zip code is required.”, “Input missing”
Confirmation message	Count 1	“Are you sure you want to delete customer?”, “Please wait while searching,”, “Processing complete”

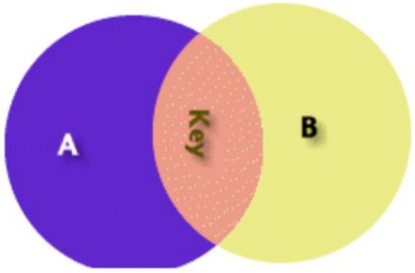
Counting FTR and RET to Measure Function Size

- File Type Referenced (FTR) — For size of EI, EO, EQ
  - File type referenced by a transaction. An FTR must also be an ILF or EIF
- Record Element Type (RET) — For size of ILF, EIF
  - User recognizable subgroup of data elements within an ILF or EIF. RETs within ILF are logically related with a parent-child relationship (one-to-many). The child information is a subset of the parent information and is meaningless when not linked to the parent.
  - E.g. An employee file consists of subgroups of bio data, salary data, and promotion data
  - E.g. An employee data consists of either salaried employee data or hourly employee data, and dependent data (optional)
  - E.g. A supplier database has supplier table, supplier address(es) table, and supplier phone(s) table

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Counting RET (1 of 2)



Some A are B

2 ILFs, 1 RET each

Customer Table

Customer Number	Name	Address	City	State	Zip Code	Balance	Credit Limit	Sales Rep Number
AN91	Atwater Nelson	215 Watkins	Oakdale	IN	48101	$347	$700	04
AW52	Alliance West	266 Ralston	Allanson	IN	48102	$49	$400	07
BD22	Betodial	542 Prairie	Oakdale	IN	48101	$57	$400	07
CE76	Carson Enterprise	96 Prospect	Bishop	IL	61354	$425	$900	11

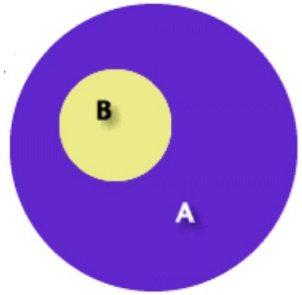
Sales Representative Table

Sales Rep Number	Last Name	First Name	Address	City	State	Zip Code	Area Manager Number
04	Right	Mike	95 Stockton	Oakdale	IN	48101	14
05	Perry	Tom	198 Pearl	Oakdale	IN	48101	17
07	Sanchez	Rachel	867 Bedford	Benson	MI	49246	17
11	Morris	Katie	96 Prospect	Bishop	IL	61354	21

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Counting RET (2 of 2)



All B are A

1 ILF, 2 RETs

Album database has Album and Songs tables.
Album has fields: artist, album name, publication date, and songs.
Key is both artist and album name
Songs has subfields: track number, song name, and length.

1. Born to Run (4:30)
2. Thunder Road (4:48)
3. Badlands (4:02)
4. River (5:00)
5. Hungry Heart (3:20)
6. Atlantic City (3:56)
7. Dancing in the Dark (4:03)
8. Born in the U.S.A. (4:41)
9. My Hometown (4:12)
10. Glory Days (3:49)
11. Brilliant Disguise (4:15)
12. Human Touch (5:10)
13. Better Days (3:44)
14. Streets of Philadelphia (3:16)
15. Murder Incorporated (4:27)
16. Secret Garden (3:57)
17. Blood Brothers (4:34)
18. This Hard Land (4:51)



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Rating FPA Components

Transaction: count FTR and DET

Data: count RET and DET

EI

Files Type Referenced (FTR)	Data Elements (DET's)		
	1-4	5-15	Greater than 15
Less than 2	Low (3)	Low (3)	Average (4)
2	Low (3)	Average (4)	High (6)
Greater than 2	Average (4)	High (6)	High (6)

ILF

Record Element Types (RET)	Data Elements		
	1 to 19	20 - 50	51 or More
1 RET	Low (7)	Low(7)	Average (10)
2 to 5 RET	Low (7)	Average (10)	High (15)
6 or More RET	Average (10)	High (15)	High (15)

EO

File Types Referenced (FTR)	Data Elements		
	1-5	6-19	Greater than 19
less than 2	Low (4)	Low (4)	Average (5)
2 or 3	Low (4)	Average (5)	High (7)
Greater than 3	Average (5)	High (7)	High (7)

EIF

Record Element Types (RET)	Data Elements		
	1 to 19	20 - 50	51 or More
1 RET	Low (5)	Low(5)	Average (7)
2 to 5 RET	Low (5)	Average (7)	High (10)
6 or More RET	Average (7)	High (10)	High (10)

EQ

File Types Referenced (FTR)	Data Elements		
	1-5	6-19	Greater than 19
less than 2	Low (3)	Low (3)	Average (4)
2 or 3	Low (3)	Average (4)	High (6)
Greater than 3	Average (4)	High (6)	High (6)

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EI Size Example

- **DET**: 12

  1. Customer
  2. Contact
  3. Alt. Contact
  4. Bill to
  5. Phone
  6. Fax
  7. Alt. Phone
  8. Ship to
  9. Type
  10. Button (OK)
  11. Error message ("Customer info incomplete")
  12. Save confirmation message ("Successfully saved")

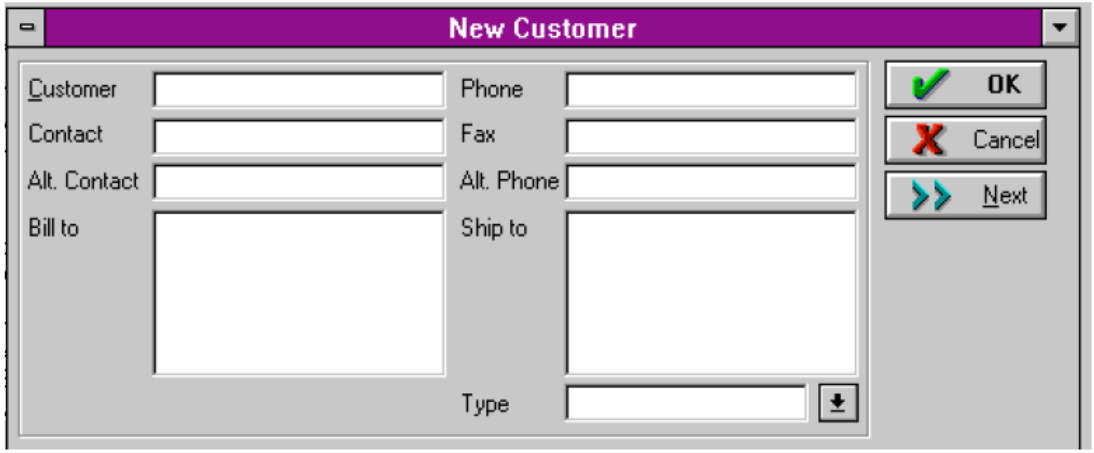
- **FTR**: 1

  1. Customer data

- **Rating**: Low
- **Value**: 3

Assume OK saves the new customer data.

Cancel and Next have no effect on input new customer data and are not counted.

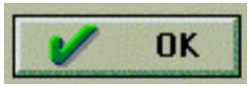


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EO Size Example

Assume that user roles restrict what information is displayed on a monthly sales report; user role needs to be checked first.

Print Monthly Sales



Ice cream sales data include flavor, month, and sales.

Flavor	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Vanilla	80	85	85	90	110	120	135	145	90	84	75	70	1169
Chocolate	75	80	70	83	100	105	109	120	80	70	69	65	1026
Strawberry	30	35	35	40	70	80	95	105	40	34	25	20	609
Pistachio	8	9	9	9	11	12	14	15	9	8	8	7	119
Other	12	13	13	13	15	17	19	20	14	13	13	12	174
Total	205	222	212	235	306	334	372	405	233	209	190	174	

- **DET**: 7
  1. OK button (input)
  2. User credential (input)
  3. Flavor (output, recursive)
  4. Month (output, recursive)
  5. Sales (output, recursive)
  6. Total by flavor (output, derived, recursive)
  7. Total by month (output, derived, recursive)
- **FTR**: 2
  1. Ice cream sales data
  2. Security file (access control)
- **Rating**: Average
- **Value**: 5

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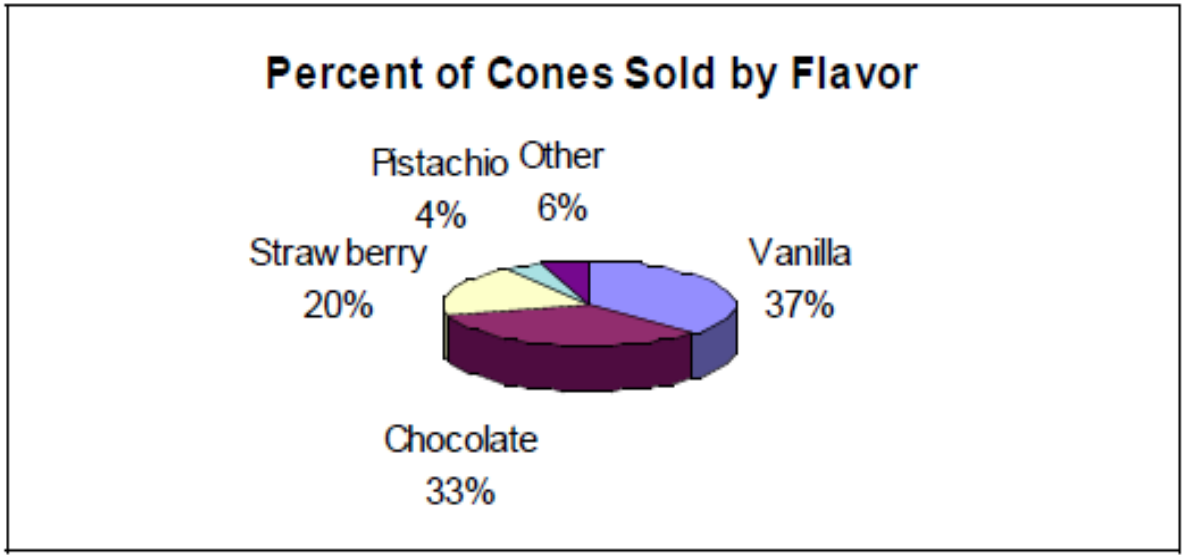
EO Size Example

- **DET**: 3
  1. OK button (input)
  2. Flavor (output, recursive)
  3. % of cones sold (output, derived, recursive)
- **FTR**: 1
  1. Ice cream sales data
- **Rating**: Low
- **Value**: 4

Ice cream sales data include flavor and no. of cones sold.

Print Sales Percent

OK



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EQ Size Example

- **DET**: 4
  1. State drop down list and map (input, multiple action keys)
  2. Go button (input)
  3. ~~State (output, repetitive)~~
  4. Population (output)
  5. % of growth (output)
- **FTR**: 1
  1. US census data
- **Rating**: Low
- **Value**: 3

DET is dynamic.

US census data include state, no. of population, and % of growth.

To begin, select a state from this list or use the map to the right.

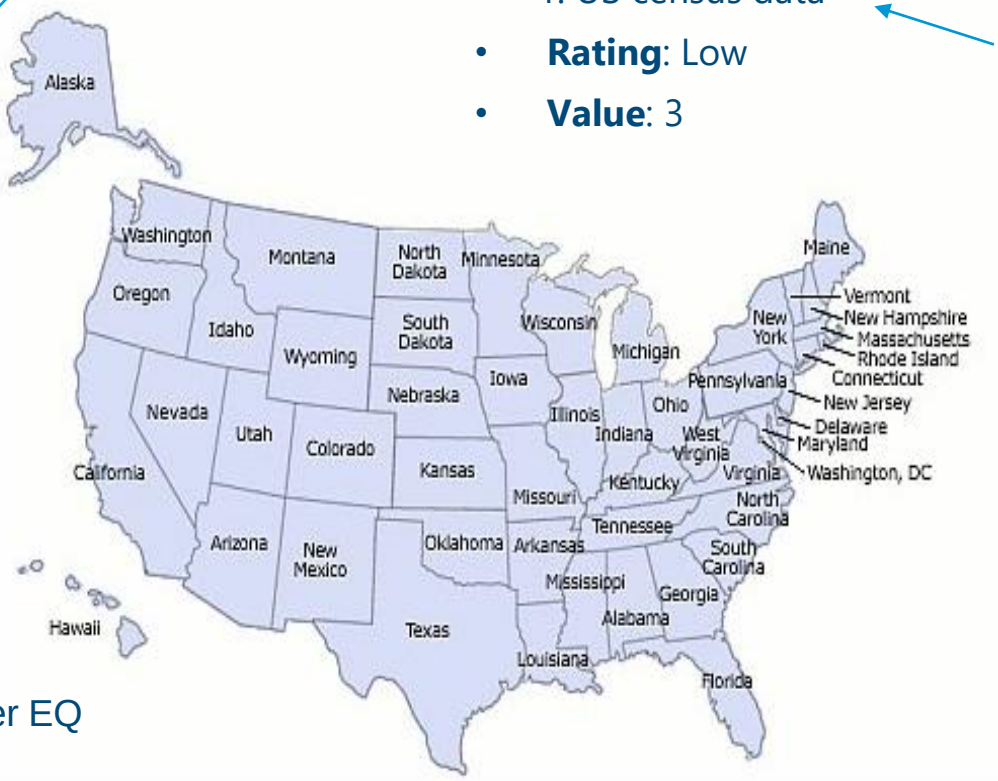
Alabama Go

The population of Alabama is 4,452,000 people (April 1, 2000). The population has grown 9.9% since 1990.

View more USA QuickFacts.

See our thematic maps

Source: U.S. Census Bureau



Other EQ

e or in part, except
room use.

ILF and EIF Size Example

- ILF
  - A customer database consists of customer table, contact(s) table, and credit card(s) table (i.e. in this case, one table as one subgroup of data which is dependent on a parent-child relationship.) There are 18 unique data elements.
  - **RET**: 3
    1. Customer table
    2. Contact(s) table
    3. Credit card(s) table
  - **DET**: 18
  - **Rating**: Low
  - **Value**: 7

- EIF
  - The application exchanges tax data with a government tax data service, involving 20 unique data elements.
  - **RET**: 1
    1. Tax data
  - **DET**: 20
  - **Rating**: Low
  - **Value**: 5

*ILF, EIF, and RET are logical, and not necessarily tied to physical structure of data like database and table

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Calculating Function Points

Counts of transaction or data

Type of Component	Complexity of Components			
	Low	Average	High	Total
External Inputs	2 x 3 = ___	___ x 4 = ___	___ x 6 = ___	6
External Outputs	3 x 4 = ___	1 x 5 = ___	___ x 7 = ___	17
External Inquiries	2 x 3 = ___	___ x 4 = ___	___ x 6 = ___	6
Internal Logical Files	3 x 7 = ___	___ x 10 = ___	___ x 15 = ___	21
External Interface Files	1 x 5 = ___	___ x 7 = ___	___ x 10 = ___	5

Unadjusted FP	Total Number of Unadjusted Function Points	55
VAF	Multiplied Value Adjustment Factor	___
Adjusted FP	Total Adjusted Function Points	___

https://www.softwaremetrics.com/fpafund.htm

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Calculating Adjusted Function Points

- Value Adjustment Factor (VAF) is based on 14 general system characteristics (GSCs) that rate the general functionality of the software
  - In the ISO standard for the IFPUG technique (ISO/IEC 20926), this step is optional
- Each characteristic has associated descriptions that help determine its degree of influence
  - 0: No influence
  - 1: Incidental
  - 2: Moderate
  - 3: Average
  - 4: Significant
  - 5. Essential

$$VAF = 0.65 + 0.01 * \sum_{i=1}^{14} F_i \; ; \; VAF \; is \; in \; [0.65, 1.35]$$

$$Adjusted \; FP = Unadjusted \; FP * VAF$$

How to determine degree of influence (Fi) pp.57-71
https://www.slideshare.net/arafkarsh/function-point-analysis-65711721

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General System Characteristic (Fi)		Brief Description
1.	Data communications	How many communication facilities are there to aid in the transfer or exchange of information with the application or system?
2.	Distributed data processing	How are distributed data and processing functions handled?
3.	Performance	Was response time or throughput required by the user?
4.	Heavily used configuration	How heavily used is the current hardware platform where the application will be executed?
5.	Transaction rate	How frequently are transactions executed daily, weekly, monthly, etc.?
6.	On-Line data entry	What percentage of the information is entered On-Line?
7.	End-user efficiency	Was the application designed for end-user efficiency?
8.	On-Line update	How many ILF's are updated by On-Line transaction?
9.	Complex processing	Does the application have extensive logical or mathematical processing?
10.	Reusability	Was the application developed to meet one or many user's needs?
11.	Installation ease	How difficult is conversion and installation?
12.	Operational ease	How effective and/or automated are start-up, back-up, and recovery procedures?
13.	Multiple sites	Was the application specifically designed, developed, and supported to be installed at multiple sites for multiple organizations?
14.	Facilitate change	Was the application specifically designed, developed, and supported to facilitate change?

Labor Estimate by Function Points

- Estimated total function points of the project is 55 function points.
- Average productivity is 0.25 function points per person-hour (ph).
- Estimated effort = 55 function points / 0.25 function points per ph = 220 ph.
- Average labor rate per person is $50 per hour.
- Estimated labor cost = 220 ph * $50 per hour = $11,000.

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Determining KLOC from FP

- The process is called backfiring
- See full list of gearing factors for languages
  - https://www.qsm.com/resources/function-point-languages-table
  - They were drawn from 2,192 completed function points projects. (Only single-language projects are used.)

$$KLOC_{lang} = (FP * gearing_factor_{lang}) / 1000$$

  - E.g. Average gearing factor for Java is 53 LOC/FP, LOC of Java program of 27 FPs is 27*53/1000 = 1.431 KLOC
- Another way to estimate labor cost from FP is to use KLOC with COCOMO II model.

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COCOMO (1 of 2)

- COCOMO (Constructive Cost Estimation Model) was first proposed by Barry Boehm in 1981
- One of the most generally used software estimation models (parametric estimating)
- Predicts the efforts and schedule of a software product
- First version is COCOMO 81
  - Database of 63 projects
  - Statistical model building (fitting actual data to equation)
  - Can be calibrated based on company-specific historical data
  - Designed for the waterfall model, which was largely superseded by incremental iterative methods
  - Database became outdated by new tools, languages, and practices

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COCOMO (2 of 2)

- Latest version is COCOMO II.2000
  - Database of 161 projects
  - Accounts for incremental iterative methods and software built using off-the-shelf or reusable components

COCOMO II Model Definition Manual version 2.1, 2000

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COCOMO II Models (1 of 2)

- Early design model is used after requirement and early design phase
- Post-architecture model is used after system design phase
- Models are functions of size, scale factors, and effort multipliers
- Size estimate (KSLOC) is by
  - Unadjusted function points (UFP) converted to SLOC and then KSLOC, historical data,or expert opinion

COCOMO II Model Definition Manual version 2.1, 2000
Boehm et al. Cost Models for Future Software Life Cycle Processes: COCOMO 2.0, Annals of Software Engineering 1(1995).

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COCOMO II Models (2 of 2)

- A person-month is the amount of time (effort) that one person spends working on the software development project for one month.
  - The number of person-days per person-month is 19 (excluding average monthly time off because of holidays, vacation, and sick leave)
- To convert a COCOMO estimate in person-months to other units:
  - Person-hours: multiply by 152 (19*8)
  - Person-days: multiply by 19
  - Person-years: divide by 12

COCOMO II Model Definition Manual version 2.1, 2000
https://ase.in.tum.de/repos/oose/trunk/lectures/L32_Estimation_ch13lect3.ppt

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Early Design Model and Post-architecture Model

- PM is the amount of effort in person-months
  - A = Effort coefficient, E = Exponential scale factor, EM = Effort multipliers
  - A and B can be calibrated to organization's own database of projects.

$$PM = A * Size^{E} * \prod_{i=1}^{n} EM_i$$

$$where\ A = 2.94$$

$$E = B + 0.01 * \sum_{j=1}^{5} SF_j \quad where\ B = 0.91$$

$$n = 7\ for\ Early\ Design\ Model \quad or \quad n = 17\ for\ Post-architecture\ Model$$

COCOMO II Model Definition Manual version 2.1, 2000

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5 Scale Factors (1 of 3)

- Precedentedness (PREC)
  - If a product is similar to several previously developed projects, then the precedentedness is high.
- Development Flexibility (FLEX)
  - Needs for conformance with pre-established requirements and external interface specifications
- Architecture / Risk Resolution (RESL)
  - Risk management plan; established schedule, budget, architecture; identification of risk items and criticality; tool support available for resolving risk items; uncertainty in key architecture drivers
- Team Cohesion (TEAM)
  - Differences in stakeholder objectives and cultures; difficulties in reconciling objectives; stakeholders' lack of experience and familiarity in operating as a team
- Process Maturity (PMAT)
  - Evaluation based on Capability Maturity Model (CMM)

- See COCOMO II Model Definition Manual version 2.1, 2000 pp. 18-24, for how to qualitatively rate scale factors

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Table 10. Scale Factor Values, SF_j, for COCOMO II Models

Extra High rating for all SFs is 0.00

Scale Factors	Very Low	Low	Nominal	High	Very High	Extra High
PREC	thoroughly unprecedented	largely unprecedented	somewhat unprecedented	generally familiar	largely familiar	thoroughly familiar
SF_j:	6.20	4.96	3.72	2.48	1.24	0.00
FLEX	rigorous	occasional relaxation	some relaxation	general conformity	some conformity	general goals
SF_j:	5.07	4.05	3.04	2.03	1.01	0.00
RESL	little (20%)	some (40%)	often (60%)	generally (75%)	mostly (90%)	full (100%)
SF_j:	7.07	5.65	4.24	2.83	1.41	0.00
TEAM	very difficult interactions	some difficult interactions	basically cooperative interactions	largely cooperative	highly cooperative	seamless interactions
SF_j:	5.48	4.38	3.29	2.19	1.10	0.00
PMAT	The estimated Equivalent Process Maturity Level (EPML) or					
	SW-CMM Level 1 Lower	SW-CMM Level 1 Upper	SW-CMM Level 2	SW-CMM Level 3	SW-CMM Level 4	SW-CMM Level 5
SF_j:	7.80	6.24	4.68	3.12	1.56	0.00

5 Scale Factors (3 of 3)

- Process Maturity

Table 15. PMAT Ratings for Estimated Process Maturity Level (EPML)

PMAT Rating	Maturity Level	EPML
Very Low	CMM Level 1 (lower half)	0
Low	CMM Level 1 (upper half)	1
Nominal	CMM Level 2	2
High	CMM Level 3	3
Very High	CMM Level 4	4
Extra High	CMM Level 5	5

18 Key Process Areas (KPA)*	KPA Compliance Levels	KPA Compliance Percentage**	EPML
Requirements Management Software Project Planning Software Project Tracking and Oversight Software Subcontract Management Software Quality Assurance (SQA) Software Configuration Management (SCM) Organization Process Focus Organization Process Definition Training Program Integrated Software Management Software Product Engineering Intergroup Coordination Peer Reviews Quantitative Process Management Software Quality Management Defect Prevention Technology Change Management Process Change Management	Almost Always	100	$EPML = 5 * \left(\sum_{i=1}^{n} \frac{KPA\%}{100} \right) * \frac{1}{n}$
	Frequently	75	
	About Half	50	
	Occasionally	25	
	Rarely If Ever	1	
	Does Not Apply	Not counted***	
	Don't Know	Not counted***	* See COCOMO II Model Definition Manual version 2.1, 2000, pp. 20-24 ** Otherwise, use compliance percentage from CMM assessment *** Making n less than 18

17 Effort Multipliers of Post-architecture Model Cost Drivers (1 of 6)

- Product factors
  - Required Software Reliability (RELY)
  - Data Base Size (DATA)
  - Product Complexity (CPLX)
  - Developed for Reusability (RUSE)
  - Documentation Match to Life-Cycle Needs (DOCU)
- Platform factors
  - Execution Time Constraint (TIME)
  - Main Storage Constraint (STOR)
  - Platform Volatility (PVOL)
- Personnel factors
  - Analyst Capability (ACAP)
  - Programmer Capability (PCAP)
  - Personnel Continuity (PCON)
  - Applications Experience (APEX)
  - Platform Experience (PLEX)
  - Language and Tool Experience (LTEX)
- Project factors
  - Use of Software Tools (TOOL)
  - Multisite Development (SITE)
  - Required Development Schedule (SCED)

When assessment of cost driver falls between rating levels, round toward Nominal.

COCOMO II Model Definition Manual version 2.1, 2000, pp. 25-35

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17 Effort Multipliers of Post-architecture Model Cost Drivers (2 of 6)

- Required Software Reliability (RELY)

Nominal rating for all EMs is 1.00

RELY Descriptors:	slight inconven-ience	low, easily recoverable losses	moderate, easily recoverable losses	high financial loss	risk to human life	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	0.82	0.92	1.00	1.10	1.26	n/a

- Data Base Size (DATA)

DATA* Descriptors		Testing DB bytes/Pgm SLOC < 10	10 ≤ D/P < 100	100 ≤ D/P < 1000	D/P ≥ 1000	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	n/a	0.90	1.00	1.14	1.28	n/a

* DATA is rated as Low if D/P is less than 10 and it is very high if it is greater than 1000. P is measured in equivalent source lines of code (SLOC), which may involve function point or reuse conversions.

- Product Complexity (CPLX)
  - Five areas: control operations, computational operations, device-dependent operations, data management operations, and user interface management operations.
  - See COCOMO II Model Definition Manual version 2.1, 2000, for how to qualitatively rate product complexity

Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	0.73	0.87	1.00	1.17	1.34	1.74

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17 Effort Multipliers of Post-architecture Model Cost Drivers (3 of 6)

- Developed for Reusability (RUSE)

RUSE Descriptors:		none	across project	across program	across product line	across multiple product lines
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	n/a	0.95	1.00	1.07	1.15	1.24

- Documentation Match to Life-Cycle Needs (DOCU)

DOCU Descriptors:	Many life-cycle needs uncovered	Some life-cycle needs uncovered.	Right-sized to life-cycle needs	Excessive for life-cycle needs	Very excessive for life-cycle needs	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	0.81	0.91	1.00	1.11	1.23	n/a

- Execution Time Constraint (TIME)

TIME Descriptors:			≤ 50% use of available execution time	70% use of available execution time	85% use of available execution time	95% use of available execution time
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	n/a	n/a	1.00	1.11	1.29	1.63

- Main Storage Constraint (STOR)

STOR Descriptors:			≤ 50% use of available storage	70% use of available storage	85% use of available storage	95% use of available storage
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	n/a	n/a	1.00	1.05	1.17	1.46

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17 Effort Multipliers of Post-architecture Model Cost Drivers (4 of 6)

- Platform Volatility (PVOL)

PVOL Descriptors:		Major change every 12 mo.; Minor change every 1 mo.	Major: 6 mo.; Minor: 2 wk.	Major: 2 mo.;Minor: 1 wk.	Major: 2 wk.;Minor: 2 days	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	n/a	0.87	1.00	1.15	1.30	n/a

- Analyst Capability (ACAP)

ACAP Descriptors:	15th percentile	35th percentile	55th percentile	75th percentile	90th percentile	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.42	1.19	1.00	0.85	0.71	n/a

- Programmer Capability (PCAP)

PCAP Descriptors	15th percentile	35th percentile	55th percentile	75th percentile	90th percentile	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.34	1.15	1.00	0.88	0.76	n/a

- Personnel Continuity (PCON)

PCON Descriptors:	48% / year	24% / year	12% / year	6% / year	3% / year	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.29	1.12	1.00	0.90	0.81	n/a

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17 Effort Multipliers of Post-architecture Model Cost Drivers (5 of 6)

- Applications Experience (APEX)

APEX Descriptors:	≤ 2 months	6 months	1 year	3 years	6 years	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.22	1.10	1.00	0.88	0.81	n/a

- Platform Experience (PLEX)

PLEX Descriptors:	≤ 2 months	6 months	1 year	3 years	6 year	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.19	1.09	1.00	0.91	0.85	n/a

- Language and Tool Experience (LTEX)

LTEX Descriptors:	≤ 2 months	6 months	1 year	3 years	6 year	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.20	1.09	1.00	0.91	0.84	n/a

- Use of Software Tools (TOOL)

TOOL Descriptors	edit, code, debug	simple, frontend, backend CASE, little integration	basic life-cycle tools, moderately integrated	strong, mature life-cycle tools, moderately integrated	strong, mature, proactive life-cycle tools, well integrated with processes, methods, reuse	
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.17	1.09	1.00	0.90	0.78	n/a

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17 Effort Multipliers of Post-architecture Model Cost Drivers (6 of 6)

- Multisite Development (SITE)

SITE: Collocation Descriptors:	Inter-national	Multi-city and Multi-company	Multi-city or Multi-company	Same city or metro. area	Same building or complex	Fully collocated
SITE: Communications Descriptors:	Some phone, mail	Individual phone, FAX	Narrow band email	Wideband electronic communication.	Wideband elect. comm., occasional video conf.	Interactive multimedia
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.22	1.09	1.00	0.93	0.86	0.80

- Required Development Schedule (SCED)
  - Effect of schedule compression / expansion for the whole project, i.e. percentage of schedule stretch-out or acceleration with respect to a nominal schedule
  - Accelerated schedules produce more effort in the earlier phases to eliminate risks and refine the architecture, more effort in the later phases to accomplish more testing and documentation in parallel
  - Stretch-outs do not add or decrease effort, i.e. smaller team size is generally balanced by carrying project administrative functions over a longer period of time

SCED Descriptors	75% of nominal	85% of nominal	100% of nominal	130% of nominal	160% of nominal	
Rating Level	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multiplier	1.43	1.14	1.00	1.00	1.00	n/a

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7 Effort Multipliers of Early Design Model Cost Drivers (1 of 4)

- Personnel Capability (PERS)
- Product Reliability and Complexity (RCPX)
- Developed for Reusability (RUSE)
- Platform Difficulty (PDIF)
- Personnel Experience (PREX)
- Facilities (FCIL)
- Required Development Schedule (SCED)

Table 35. Early Design and Post-Architecture Effort Multipliers

Early Design Cost Driver	Counterpart Combined Post-Architecture Cost Drivers
PERS	ACAP, PCAP, PCON
RCPX	RELY, DATA, CPLX, DOCU
RUSE	RUSE
PDIF	TIME, STOR, PVOL
PREX	APEX, PLEX, LTEX
FCIL	TOOL, SITE
SCED	SCED

- Involving the sum of post-architecture rating levels, i.e. very low = 1, low = 2, nominal = 3, high = 4, very high = 5, extra high = 6

When assessment of cost driver falls between rating levels, round toward Nominal.

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7 Effort Multipliers of Early Design Model Cost Drivers (2 of 4)

- Personnel Capability (PERS)

PERS Descriptors:							
• Sum of ACAP, PCAP, PCON Ratings	3, 4	5, 6	7, 8	9	10, 11	12, 13	14, 15
• Combined ACAP and PCAP Percentile	20%	35%	45%	55%	65%	75%	85%
• Annual Personnel Turnover	45%	30%	20%	12%	9%	6%	4%
Rating Levels	Extra Low	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	2.12	1.62	1.26	1.00	0.83	0.63	0.50

- Product Reliability and Complexity (RCPX)

RCPX Descriptors:							
• Sum of RELY, DATA, CPLX, DOCU Ratings	5, 6	7, 8	9 - 11	12	13 - 15	16 - 18	19 - 21
• Emphasis on reliability, documentation	Very Little	Little	Some	Basic	Strong	Very Strong	Extreme
• Product complexity	Very simple	Simple	Some	Moderate	Complex	Very complex	Extremely complex
• Database size	Small	Small	Small	Moderate	Large	Very Large	Very Large
Rating Levels	Extra Low	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	0.49	0.60	0.83	1.00	1.33	1.91	2.72

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7 Effort Multipliers of Early Design Model Cost Drivers (3 of 4)

- Developed for Reusability (RUSE) same as post-architecture model

RUSE Descriptors:		none	across project	across program	across product line	across multiple product lines
Rating Levels	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	n/a	0.95	1.00	1.07	1.15	1.24

- Platform Difficulty (PDIF)

PDIF Descriptors:					
• Sum of TIME, STOR, and PVOL ratings	8	9	10 - 12	13 - 15	16, 17
• Time and storage constraint	≤ 50%	≤ 50%	65%	80%	90%
• Platform volatility	Very stable	Stable	Somewhat volatile	Volatile	Highly volatile
Rating Levels	Low	Nominal	High	Very High	Extra High
Effort Multipliers	0.87	1.00	1.29	1.81	2.61

- Personnel Experience (PREX)

PREX Descriptors:							
• Sum of APEX, PLEX, and LTEX ratings	3, 4	5, 6	7, 8	9	10, 11	12, 13	14, 15
• Applications, Platform, Language and Tool Experience	≤ 3 mo.	5 months	9 months	1 year	2 years	4 years	6 years
Rating Levels	Extra Low	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.59	1.33	1.22	1.00	0.87	0.74	0.62

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7 Effort Multipliers of Early Design Model Cost Drivers (4 of 4)

- Facilities (FCIL)

FCIL Descriptors:							
• Sum of TOOL and SITE ratings	2	3	4, 5	6	7, 8	9, 10	11
• TOOL support	Minimal	Some	Simple CASE tool collection	Basic life-cycle tools	Good; moderately integrated	Strong; moderately integrated	Strong; well integrated
• Multisite conditions	Weak support of complex multisite development	Some support of complex M/S devel.	Some support of moderately complex M/S devel.	Basic support of moderately complex M/S devel.	Strong support of moderately complex M/S devel.	Strong support of simple M/S devel.	Very strong support of collocated or simple M/S devel.
Rating Levels	Extra Low	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multipliers	1.43	1.30	1.10	1.0	0.87	0.73	0.62

- Required Development Schedule (SCED) same as post-architecture model

SCED Descriptors	75% of nominal	85% of nominal	100% of nominal	130% of nominal	160% of nominal	
Rating Level	Very Low	Low	Nominal	High	Very High	Extra High
Effort Multiplier	1.43	1.14	1.00	1.00	1.00	n/a

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Calendar Time to Develop

- TDEV is the amount of calendar time, in months, to develop the product
  - From determination of a product's requirements baseline to completion of an acceptance activity
  - PM_{NS} = PM with nominal schedule, i.e. excluding SCED effort multiplier
  - C = TDEV coefficient, D = TDEV scaling base-exponent
  - SCED% = compression/expansion percentage in the SCED effort multiplier rating
  - A (in PM_{NS}), B, C, and D can be calibrated to organization's own database of projects. (It is recommended that at least A and C is calibrated to the local development environment.)

$$TDEV = \left[C * (PM_{NS})^{D+0.2*(E-B)} \right] * \frac{SCED\%}{100}$$

$$where\ C = 3.67,\ D = 0.28, B = 0.91$$

$$E = B + 0.01 * \sum_{j=1}^{5} SF_j \quad (same\ as\ PM\ calculation)$$

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Staff Size, Productivity, and Cost

$$Staff\ Size = \frac{PM}{TDEV} \quad persons$$

$$Productivity = \frac{KLOC}{PM} \quad KLOC\ /\ person-month$$

$$Development\ Labor\ Cost = PM * Average\ Labor\ Cost\ Per\ Person-Month$$

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https://www.javatpoint.com/cocomo-model

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Distribution of Effort across Activities

- Distribution from COCOMO II experiment (Waterfall), e.g.
  - A development project effort is 120 person-months, size is 8 KSLOC, and an exponent E is 1.20
    - Requirements Analysis consumes 4% of the effort, i.e. or 4.8 person-months
    - Product Design consumes 12% of the effort, i.e. 14.4 person-months
    - Programming will consume 43% of the effort, i.e. 51.6 person-months
    - And so on

Table 55. Development Activity Distribution

	Size Exponent													
	E = 1.05				E = 1.12					E = 1.20				
Size:	**S**	**I**	**M**	**L**	**S**	**I**	**M**	**L**	**VL**	**S**	**I**	**M**	**L**	**VL**
Requirements Analysis	6				5	5	5	5	5	4	4	4	4	4
Product Design	14				13	13	13	13	13	12	12	12	12	12
Programming	48	47	46	45	45	45	44.5	44.5	44.5	42	43	43	44	45
Test Planning	4				4	4	4.5	5	5.5	4	4	5	6	7
V&V	10	11	12	13	11	12	13	13.5	14	12	13	14	14	14
Project Office	7				8.5	8	7.5	7	6.5	10	9	8	7	6
CM/QA	5				6.5	6	6	6	5.5	8	7	7	7	6
Manuals	6				7	7	6.6	6	6	8	8	7	6	6

S: 2 KSLOC; I: 8 KSLOC; M: 32 KSLOC; L: 128 KSLOC; VL: 512 KSLOC

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3. Determining the Budget (1 of 3)

- Budgeting involves allocating the project cost estimate to individual work items over time
  - Material resources or work items are based on the activities in the WBS for the project
- Important goal is to produce a cost baseline
  - Time-phased budget that project managers use to measure and monitor cost performance
  - To arrange money for work each month
- Project budget = cost baseline + management reserve

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Determining the Budget (2 of 3)

Surveyor Pro Project Cost Baseline Created October 10*

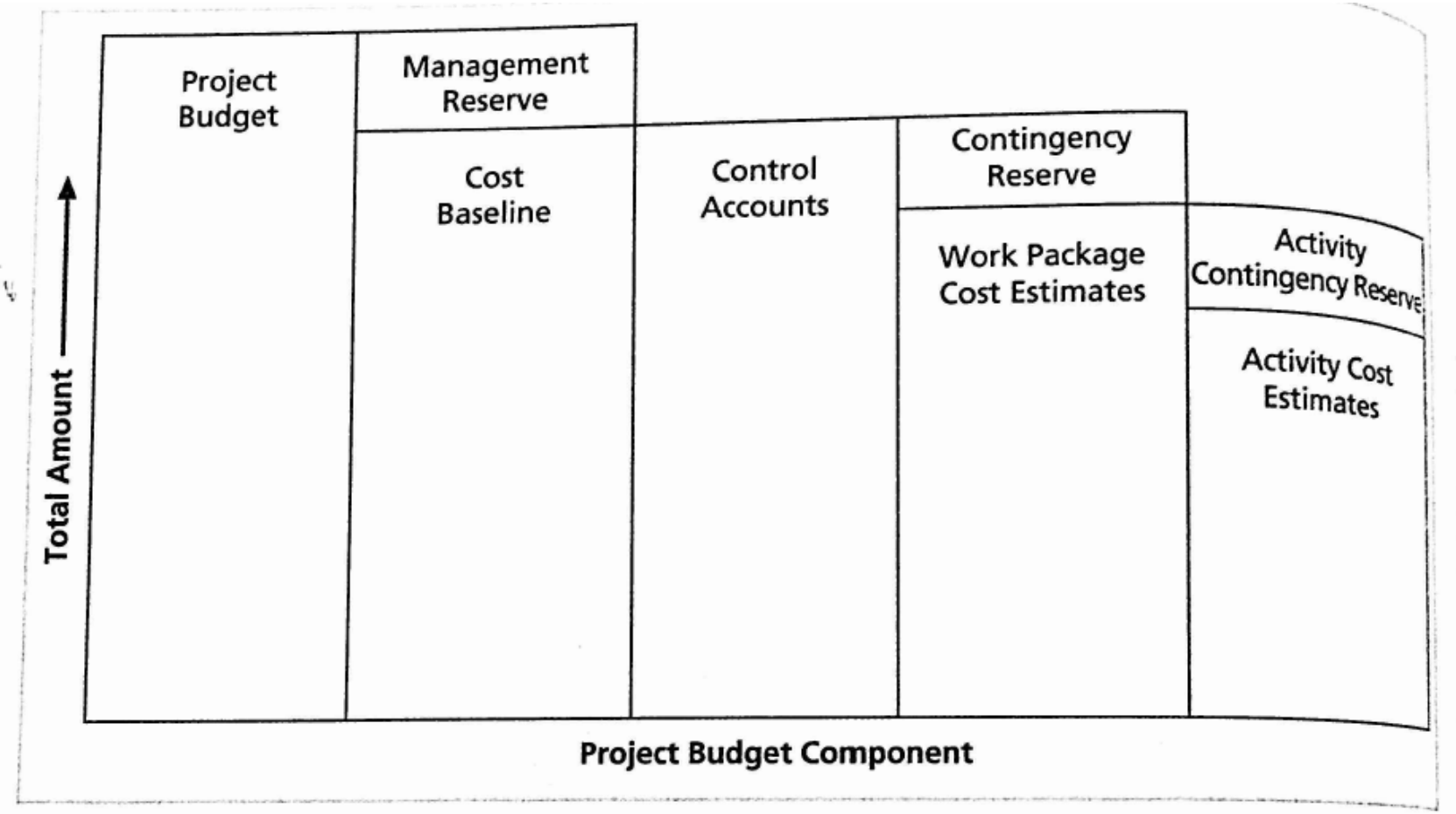
	Months												
WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
1.1 Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
1.2 Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
1.3 Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
5.1 Trainee cost									50,000				50,000
5.2 Travel cost									8,400				8,400
5.3 Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

*See the lecture slides for this chapter on the Instructor website for a larger view of this and other figures in this chapter. Numbers are rounded, so some totals appear to be off.

FIGURE 7-4 Surveyor Pro project cost baseline

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Determining the Budget (3 of 3)



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FIGURE 7-5 Project budget components

4. Controlling Costs

- Activities involved in controlling project costs
  - Monitoring cost performance
  - Ensuring that only appropriate project changes are included in a revised cost baseline
  - Informing project stakeholders of authorized changes to the project that will affect costs
- Several tools and techniques assist in project cost control
  - Expert judgment, data analysis, project management information systems, and the to-complete performance index

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Earned Value Management (EVM) (1 of 8)

- Project performance measurement technique that integrates scope, time, and cost data
  - Given a baseline (original plan plus approved changes), you can determine how well the project is meeting scope, time, and cost goals
- Earned value management involves calculating three values for each activity or summary activity from a project's WBS at time t (monitoring time)
  - Planned value (PV)
  - Actual cost (AC)
  - Earned value (EV)
- Example: a project to purchase and install a new web server
  - Planned project duration = 2 weeks
  - Planned cost = $10,000 for labor hours, hardware, and software
  - Suppose time t = end of week1,
    - Completed work = 40%
    - Actual cost = $6,000

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Earned Value Management (EVM) (2 of 8)

- Budget at completion (BAC)
  - Original total budget for the project
  - BAC = $10,000
- Planned value (PV)
  - The authorized budget assigned to scheduled work at time t
  - Expected % of work completed at time t * BAC
  - PV = $5,000 (50% * 10,000)
- Actual cost (AC)
  - Actual cost incurred for the work at time t
  - AC = $6,000
- Earned value (EV)
  - The measure of work performed at time t in terms of the budget authorized for that work
  - Actual % of work completed at time t * BAC
  - EV = $4,000 (40% * 10,000)

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Earned Value Management (EVM) (3 of 8)

- Cost variance (CV)
  - Difference between the cost actually incurred for the work and the cost that should have incurred at time t
  - CV = EV-AC
  - If CV is negative, performing the work cost more than it should be
  - If CV is positive, performing the work cost less than it should be
  - CV = $-2,000
- Schedule variance (SV)
  - Difference between the monetary value of work completed and the planned value at time t
  - SV = EV-PV
  - If SV is negative, performing the work took longer than planned (behind schedule)
  - If SV is positive, performing the work took less time than planned (ahead of schedule)
  - SV = $-1,000

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Earned Value Management (EVM) (4 of 8)

- Cost performance index (CPI)
  - How efficient the actual cost spent is, compared to the monetary value of the completed work, determined at time t
  - CPI = EV/AC
  - If CPI is 1 or 100%, the actual cost is exactly as budgeted
  - If CPI is < 1 or 100%, the actual cost is over budget
  - If CPI is > 1 or 100% the actual cost is under budget
  - CPI = $4,000/6,000 = 0.67 (66.67%) = 33.33% over budget
    - The project team would not be able to complete the project within the original budget (BAC), if it continues to perform at the current cost efficiency (CPI)
- Schedule performance index (SPI)
  - How close the work is to being completed, compared to the amount of work that should be finished
  - SPI = EV/PV
  - If SPI is 1 or 100%, the work is on schedule
  - If SPI is < 1 or 100%, the work is behind schedule
  - If SPI is > 1 or 100%, the work is ahead of schedule
  - SPI = $4,000/5,000 = 0.8 (80%) = 20% behind schedule
    - The project team would not be able to complete the project within the original planned duration, if it continues to perform at the current schedule efficiency (SPI)

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Earned Value Management (EVM) (5 of 8)

- Estimate at completion (EAC)
  - An estimated cost of completing a project based on the performance to date, determined at time t
  - EAC = BAC/CPI
  - EAC = $10,000/0.67 = $14,925
- Estimated time to complete
  - An estimated time of completing a project based on the performance to date, determined at time t
  - Estimated time to complete = Original time estimate/SPI
  - Estimated time to complete = 2 weeks/0.8 = 2.5 weeks
- Estimated to complete (ETC)
  - The amount of money to spend more to complete the project based on the performance to date, determined at time t
  - ETC = EAC-AC
  - ETC = $14,925-6,000 = $8,925

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Earned Value Management (EVM) (6 of 8)

- To-complete performance index (TCPI)
  - The future cost performance index (future CPI) that you must follow for the remaining work if you want to complete the work within the given budget, i.e. BAC or EAC
  - TCPI = monetary value of remaining work/remaining funds
    - TCPI < 1 : project has more funds and less work, and it is easier to complete the project
    - TCPI = 1 : project has just enough funds to complete the work
    - TCPI > 1 : project has less funds and more work, and it is difficult to complete the project
  - To meet the BAC (i.e. to complete on original budget)
    - TCPI = (BAC-EV)/(BAC-AC)
    - TCPI = ($10,000-4,000)/($10,000-6,000) = $6,000/4,000 = 1.5
    - If the project team wants to complete the project within original budget (BAC), the future cost efficiency should be 1.5 (123.88% improvement from current CPI)
  - If it is clear that the BAC is not obtainable, project manager/team should develop a new budget forecast (EAC), with sponsor approval, by re-estimating the cost of the remaining work (ETC)

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Earned Value Management (EVM) (7 of 8)

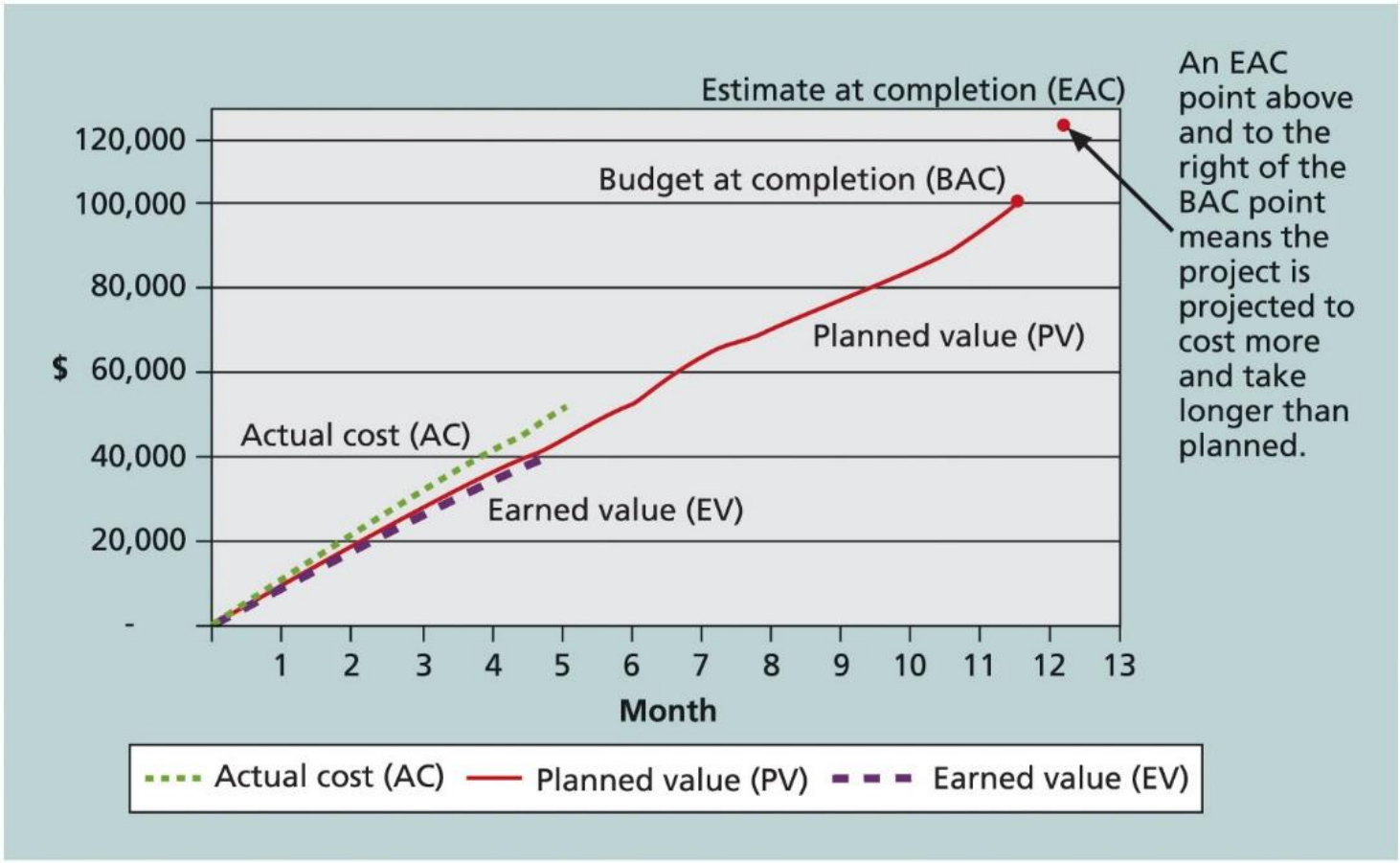
- To-complete performance index (TCPI) (cont.)
  - If the sponsor is happy with the current cost performance (CPI), to meet the new budget EAC (slide 76)
    - TCPI = (BAC-EV)/(EAC-AC)
    - TCPI = ($10,000-4,000)/($14,925-6,000) = $6,000/8,925 = 0.67
  - If the sponsor is not happy with the current cost performance (CPI) and has imposed a new budget, to meet the new budget EAC
    - EAC = $12,000
    - ETC = EAC-AC = $12,000-6,000 = $6,000
    - TCPI = (BAC-EV)/(EAC-AC)
    - TCPI = ($10,000-4,000)/($12,000-6,000) = $6,000/6,000 = 1
    - If the project team wants to complete the project within the revised budget (EAC), the future cost efficiency should be 1 (49.25% improvement from current CPI)
  - TCPI is compared with CPI to determine if a target EAC is reasonable, e.g.
    - Reasonable: if TCPI differs from CPI within 0.5 of CPI
    - Overly Optimistic: if TCPI differs from CPI more than 0.5 of CPI
    - Overly Conservative: if CPI differs from TCPI more than 0.5 of TCPI
    - Anyway, it is still a judgment call

https://acqnotes.com/acqnote/tasks/to-complete-performance-indexev

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Earned Value Management (EVM) (8 of 8)

- You can graph earned value information to track project performance
- Example of a 1-year $100,000 project after 5 months



CPI = 81.761%

EAC = $122,308

SPI = 94.203%

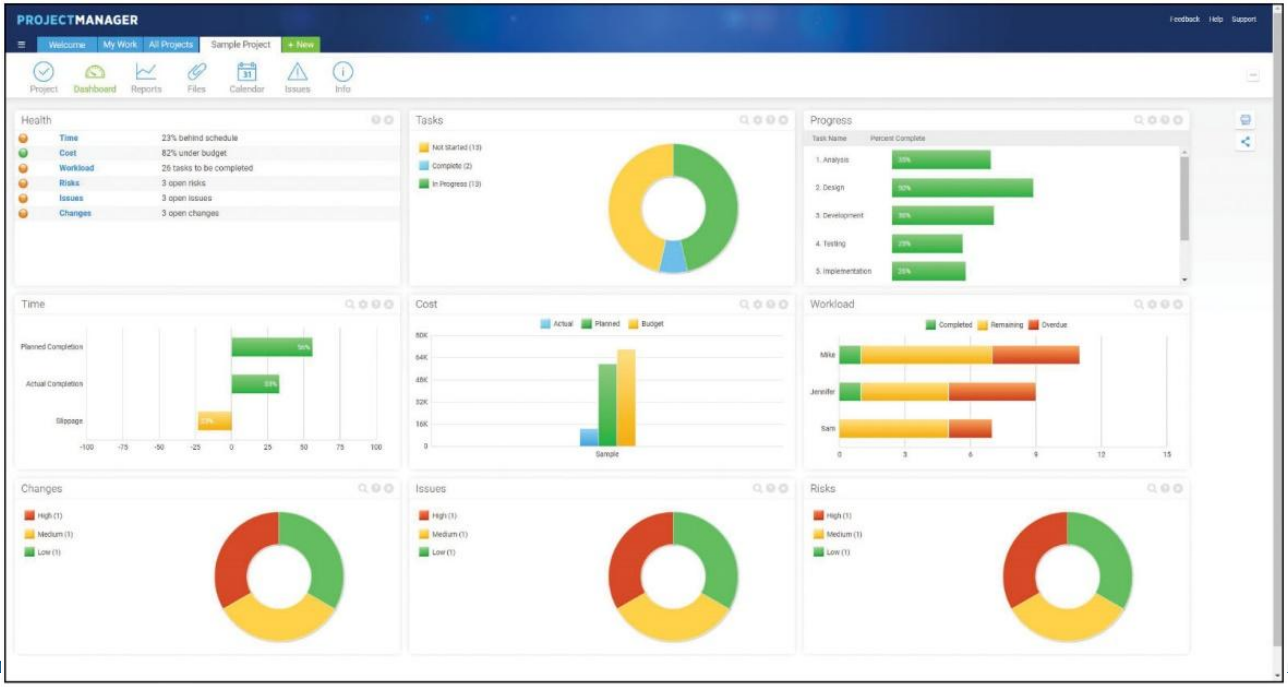
Estimated time to complete = 12.74 months

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FIGURE 7-6 Earned value chart for project after five months

Project Portfolio Management

- Many organizations collect and control an entire suite of projects or investments as one set of interrelated activities in a portfolio
- Software tools provide graphics to summarize performance on a portfolio of projects, such as cost performance



Source: www.projectmanager.com

FIGURE 4-8 Sample portfolio management software screens

Information Tech … part, except for use as permitt … use.

Using Project Management Software to Assist in Project Cost Management

- Spreadsheets are a common tool for resource planning, cost estimating, cost budgeting, and cost control
  - Many companies use more sophisticated and centralized financial applications software for cost information
- Project management software can increase a project manager's effectiveness during each process of project cost management
  - Assign costs to resources and tasks
  - Prepare cost estimates
  - Develop cost budgets
  - Monitor cost performance
  - Issue cost reports, e.g. cash flow, budget, over-budget tasks, over-budget resources, earned value reports

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Considerations for Agile/Adaptive Environments

- AgileEVM is an adapted implementation of EVM
  - Uses the Scrum framework artifacts as inputs, uses traditional EVM calculations, and is expressed in traditional EVM metrics
  - Requires a minimal set of input parameters
    - Actual cost of a project, an estimated product backlog, a release plan that provides information on the number of iterations in the release and the assumed velocity
  - All estimates can be in hours, story-points, team days, or any other consistent estimate of size
    - The critical factor is that it must be a numerical estimate of some kind

https://www.methodsandtools.com/archive/archive.php?id=61

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Chapter Summary

- Project cost management is a traditionally weak area of IT projects
  - Project managers must understand several basic principles of cost management to be effective in managing project costs
- Main processes
  - Plan cost management
  - Estimate costs
  - Determine the budget
  - Control costs
- Several software products can assist with project cost management

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